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Algorithm appreciation: People prefer algorithmic to human judgment

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Abstract

Even though computational algorithms often outperform human judgment, received wisdom suggests that people may be skeptical of relying on them (Dawes, 1979). Counter to this notion, results from six experiments show that lay people adhere *more* to advice when they think it comes from an algorithm than from a person. People showed this effect, what we call *algorithm appreciation*, when making numeric estimates about a visual stimulus (Experiment 1A) and forecasts about the popularity of songs and romantic attraction (Experiments 1B and 1C). Yet, researchers predicted the opposite result (Experiment 1D). Algorithm appreciation persisted when advice appeared jointly or separately (Experiment 2). However, algorithm appreciation waned when: people chose between an algorithm's estimate and their own (versus an external advisor's; Experiment 3) and they had expertise in forecasting (Experiment 4). Paradoxically, experienced professionals, who make forecasts on a regular basis, relied less on algorithmic advice than lay people did, which hurt their accuracy. These results shed light on the important question of when people rely on algorithmic advice over advice from people and have implications for the use of "big data" and algorithmic advice it generates.

Introduction

Although people often receive advice from other people, the rise of "big data" has increased both the availability and utility of a new source of advice: algorithms. The superior accuracy of algorithmic judgment relative to human judgment (Dawes, Faust, & Meehl, 1989) has led organizations to invest in the power of algorithms – scripts for mathematical calculations – to sift through data and produce insights. Companies including Johnson & Johnson and Jet Blue invest in complex algorithms to hire promising employees, track the satisfaction of current employees, and predict which employees are at risk for leaving the organization. Individuals in need of legal counsel can now use "Robo-advisors" such as DoNotPay to help them contest parking tickets or even file for political asylum. Indeed, decision-makers now rely on algorithms for personal use, in place of human secretaries, travel agents, headhunters, matchmakers, D.J.s, movie critics, cosmetologists, clothing stylists, sommeliers, and financial advisors (Siri, Google Search, and spell-check; Kayak, LinkedIn, OkCupid, Pandora, Netflix, Birchbox, Stitch Fix, Club W, and Betterment, respectively).

Such widespread reliance on algorithmic advice seems at odds with the judgment and decision-making literature which demonstrates human distrust of algorithmic output, sometimes referred to as "algorithm aversion" (Dietvorst, Simmons, & Massey, 2015).[1] This idea is so prevalent that it has been adopted by popular culture and the business press (Frick, 2015). Articles advise business leaders on how to overcome aversion to algorithms (Harrell, 2016). Companies in the business of selling algorithmic advice often go to great lengths to present it as purely human-generated. Stitch Fix, for example, uses a combination of algorithmic and human judgment to provide clothing recommendations to consumers. Yet, each shipment of clothes includes a personalized note from a stylist in order to focus consumers' attention on the human component.

In the present paper, we trace the history of research examining peoples' responses to algorithmic output and highlight boundary conditions for empirical evidence supporting algorithm aversion. We then present results showing that under conditions that apply to many decisions, and across a variety of estimation and forecasting tasks, people actually prefer advice from algorithms to advice from people. We call this effect "algorithm appreciation."

The first scholarly reference to psychological distrust of algorithms may belong to Meehl (1954). Importantly, it appears in the discussion section of his classic book outlining the predictive superiority of algorithms over human experts. Specifically, when researchers shared their findings with the experts in question, the conclusions were met with skepticism. It appears that the experts in the 1950s were hesitant to believe that a linear model could outperform their judgment. Similar anecdotes have circulated from other scholars (Dana and Thomas, 2006, Dawes, 1979, Hastie and Dawes, 2001). Over time, these anecdotal claims have become received wisdom in the field of judgment and decision making. In his best-selling book, *Thinking Fast and Slow*, Kahneman recounts Meehl's story: "From the very outset, clinical psychologists responded to Meehl's ideas [accuracy of algorithms] with hostility and disbelief" (2013, pg. 227).

Decades passed before the mistrust of algorithms was empirically tested. The results did not always support the received wisdom. On the one hand, in subjective domains governed by personal taste, people relied on friends over algorithmic recommender systems for book, movie, and joke recommendations (Sinha and Swearingen, 2001, Yeomans et al., unpublished data). Participants who imagined themselves as medical patients more frequently followed a subjectively worded recommendation for an operation from a doctor than from a computer (Promberger & Baron, 2006). And an influential set of papers demonstrates that after seeing an algorithm err, people relied more on human judgment than an algorithm's (Dietvorst et al., 2015, Dzindolet et al., 2002).

On the other hand, work in computer science shows that participants considering logic problems agreed more with the same argument when it came from an "expert system" than when it came from a "human"

(Dijkstra, Liebrand, & Timminga, 1998). This preference for algorithmic output persisted, *even* after they saw the algorithm err (Dijkstra, 1999). In other studies, people outsourced their memory of information to algorithmic search engines (Sparrow et al., 2011, Wegner and Ward, 2013). In the meantime, companies continue to worry about aversion to their algorithms (Haak, 2017) from their own employees, other client organizations, and individual consumers, even while many people entrust their lives to algorithms (e.g., autopilot in aviation).

Given the richness of the decision-making landscape, it is perhaps not surprising that different studies have arrived at different results. One important feature that distinguishes prior investigations is the comparison of people's reliance on algorithmic judgment to their reliance on their own, self-generated judgments. This approach has intuitive appeal, as managers and consumers often choose whether to follow their own conclusions or rely on the output of an algorithm. However, given the evidence that people routinely discount advice, such paradigms do not answer the question of how people react to algorithms, as compared to *other advisors*. Indeed, the robust result from literature on utilization of human advice is that individuals regularly (and inaccurately) discount the advice of others when making quantitative judgments (Yaniv & Kleinberger, 2000). Research has attributed such underweighting of other's input to egocentrism (Soll & Mannes, 2011). Furthermore, an extensive literature on overconfidence repeatedly demonstrates that individuals routinely report excessive confidence in their own judgment relative to that of their peers (Gino and Moore, 2007, Logg et al., 2018, Moore and Healy, 2008, Moore et al., 2015). These literatures therefore raise the question of whether individuals insufficiently trust algorithms (relative to human advisors) or merely overly trust themselves. Thus, a direct comparison of advice utilization from an algorithm versus a human advisor would prove especially useful.

Another important concern is the quality of the advice in question. The now classic literature on clinical versus actuarial judgment has demonstrated the superior accuracy of even the simplest linear models relative to individual expert judgment (for a meta-analysis, see Grove, Zald, Lebow, Snitz, & Nelson, 2000). Although it is unlikely that research participants are aware of these research findings, they may believe that algorithmic judgment is only useful in some domains. This issue of advice quality particularly plagues studies that compare recommendations from an algorithm to recommendations from a human expert, such as a doctor (Promberger & Baron, 2006). As the normative standard for how a single expert should be weighted is unclear, it is also unclear what we should conclude from people choosing an algorithm's recommendation more or less frequently than a doctor's. Thus, the accuracy of algorithmic advice becomes important for interpreting results. In order to isolate the extent to which judgment utilization is specifically driven by the fact that the advisor happens to be an algorithm, the human and algorithmic advice should be of comparable quality.

Yet another important factor is the domain of judgment. It may be perfectly reasonable to rely on the advice of close friends rather than a "black box" algorithm when making a decision reflecting one's own personal taste (Yeomans, Shah, Mullainathan, & Kleinberg, 2018). Conversely, individuals may feel more comfortable with algorithmic advice in domains that feature a concrete, external standard of accuracy, such as investment decisions or sports predictions. Relatedly, the extent to which some domains may appear "algorithmically appropriate" may depend on the historical use of algorithms by large numbers of people. For example, most people have grown comfortable with weather forecasts from meteorological models rather than one's neighbors because meteorological models have enjoyed widespread use for decades.

Conversely, the idea of fashion advice from algorithms is still relatively new and may face greater resistance.

When understanding algorithmic utilization, the algorithms' prior performance, when made available, is useful to consider. The work of Dietvorst et al. (2015) demonstrates that when choosing between their own (or another participant's) estimate and an algorithm's estimate, participants punished the algorithm after seeing it err, while showing greater tolerance for their own mistakes (or another participant's). Importantly, because the algorithm is, on average, more accurate than a single participant, choosing the human estimate decreases judgment accuracy. However, in the control conditions of the Dietvorst et al. studies, participants chose the algorithm's judgment more frequently than they chose their own (or another person's). And although the authors consistently and clearly state that "seeing algorithms err makes people less confident in them and less likely to choose them over an inferior human forecaster" (pg. 10), other research cites the paper as demonstrating generalized algorithm aversion.[2]

The current research revisits the basic question of whether individuals distrust algorithmic advice more than human advice. In our experiments, we benchmark people's responses to advice from an algorithmic advisor against their responses to advice from a human advisor. Doing so allows us to take into account that people generally discount advice relative to their own judgment.

Importantly, we also control for the quality of the advice. Participants in different conditions receive identical numeric advice, merely labeled as produced by a human or an algorithmic source, which differentiates our work from past work. Thus, any differences we observe are not a function of advice accuracy, but merely the inferences participants might make about the source. Across experiments, we describe the human and algorithmic advice in several different ways. Thus, we rule out alternative explanations based on participants' interpretations of our descriptions (Table 1). Finally, we use a variety of contexts to examine utilization of advice across different judgment domains.

We employ the Judge Advisor System (JAS) paradigm to measure the extent to which people assimilate advice from different sources (Sniezek & Buckley, 1995). The JAS paradigm requires participants to make a judgment under uncertainty, receive input ("advice"), and then make a second, potentially revised judgment. Where appropriate, we incentivized participants' final judgments. The dependent variable, Weight on Advice (WOA), is the difference between the initial and revised judgment divided by the difference between the initial judgment and the advice. WOA of 0% occurs when a participant ignores advice and WOA of 100% occurs when a participant abandons his or her prior judgment to match the advice.

Extensive prior research has documented that when seeking to maximize judgment accuracy, a person who receives advice from a single randomly-selected individual should generally average their own judgment with the advice, as reflected in a WOA of 50% (Dawes & Corrigan, 1974; Einhorn & Hogarth, 1975). Yet, people tend to only update 30–35% on average toward advice, incurring an accuracy penalty for doing so (Liberman et al., 2012, Minson et al., 2011, Soll and Larrick, 2009). Various attempts to increase advice utilization have demonstrated the effectiveness of paying for advice (Gino, 2008); highlighting advisor expertise (Harvey and Fischer, 1997, Sniezek et al., 2004); and receiving advice generated by multiple individuals (Mannes, 2009, Minson and Mueller, 2012). Could providing advice from an algorithm also increase people's willingness to adhere to advice?

In addition to capturing advice utilization using a more precise, continuous measure, WOA also taps a somewhat different psychological phenomenon than does choosing between one's own judgment and that of an algorithm. Prior work demonstrates that people are quite attached to their intuitive judgments and prefer to follow them even while explicitly recognizing that some other judgment is more likely to be objectively correct (Woolley & Risen, 2018). This could lead people to endorse an advisor but ultimately fail to act on the advice. By contrast, WOA, especially under incentivized conditions, captures actual change in the participant's *own* judgment as a function of exposure to the advice.

In six experiments, participants make quantitative judgments under uncertainty and receive advice. In the majority of experiments, we manipulate the source of advice (human versus algorithm) and examine how much weight participants give to the advice. In Experiments 2 and 4, participants choose the source of advice they prefer. In one experiment, we ask active researchers in the field to predict how participants in one of our experiments responded to advice (Experiment 1D).

Across our experiments, we find that people consistently give more weight to equivalent advice when it is labeled as coming from an algorithmic versus human source. We call this effect *algorithm appreciation*. Yet, researchers predict the opposite; they predict algorithm aversion. Together, these results suggest that algorithm aversion is not as straightforward as prior literature suggests, nor as contemporary researchers predict.

In our experiments, we report how we determined sample sizes and all conditions. All sample sizes were determined a priori by running power analyses (most at 80% power). Where possible, we based effect size estimates on prior experiments. Where that was not possible, we conservatively assumed small effect sizes, which led us to employ larger sample sizes. Pre-registrations (including all exclusions), materials, and data are posted as a supplement online at the Open Science Framework: https://osf.io/b4mk5/↗. We pre-registered analyses (and exclusions) for Experiments 1B, 1C, 1D, 2, 3, and 4. We ran Experiment 1A before pre-registration became our standard practice.

Section snippets

Benchmarking reliance on algorithmic advice against reliance on human advice

Experiments 1A, 1B, and 1C test the extent to which people are willing to adjust their estimates in response to identical advice, depending on whether it is labeled as coming from a human versus an algorithmic advisor. To generate advice, we use one of the simplest algorithms: averaging multiple independent judgments. Doing so allows us to provide high quality advice and truthfully describe it as coming from either people or from an algorithm. We test how people respond to algorithmic versus…

Experiment 2: Joint versus separate evaluation

Experiments 1A, 1B, and 1C demonstrated that in separate evaluation of advisors, our between-subjects manipulation of the source affected advice utilization. Yet, prior work finding algorithm aversion asked participants to choose between sources (joint evaluation). As attributes are easier to evaluate in a joint evaluation, due to increased information (Bazerman et al., 1992, Bazerman et al., 1999, Hsee, 1996, Hsee et al., 1999), perhaps providing participants with a counterfactual to the…

Experiment 3: The role of the self

In the introduction, we alluded to the importance of distinguishing between the extent to which individuals might disregard *algorithmic* advice from the extent to which they might disregard advice *in general*, with a preference for their own judgment. Addressing this question requires a comparison of how advice is utilized both when it comes from human versus algorithmic sources, as well as when the human judgment is produced by the participant him or herself. Such a design necessitates that we…

Experiment 4: Decision-maker expertise

Our experiments thus far have focused on how elements of the judgment context's influence people's response to algorithmic versus human advice. Experiment 4 examines whether the expertise of the decision-maker influences algorithm appreciation. We recruited professionals whose work in the field of national security for the U.S. government made them experts in geopolitical forecasting and compared their advice utilization with the diverse online sample available through Amazon's Mechanical Turk. …

General discussion

Counter to the widespread conclusion that people distrust algorithms, our results suggest that people readily rely on algorithmic advice. Our participants relied more on identical advice when they thought it came from an algorithm than from other people. They displayed this algorithm appreciation when making visual estimates and when predicting: geopolitical and business events, the popularity of songs, and romantic attraction. Additionally, they chose algorithmic judgment over human judgment…

Conclusion

Technological advances make it possible for us to collect and utilize vast amounts of data like never before. "Big data" is changing the way organizations function and communicate, both internally and externally. Organizations use algorithms to hire and fire people (Ritchel, 2013), manage employees' priorities (Copeland & Hope, 2016), and help employees choose health care plans (Silverman, 2015). Companies such as Stitch Fix use algorithms to provide clients with clothing recommendations (…

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